

May 19, 2024

CPUC Energy Division Tariff Unit
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Re: Response of the Vehicle-Grid Integration Council to Advice Letter 5536-E of Southern California Edison

Dear Sir or Madam:

Pursuant to the provisions of General Order 96-B, the Vehicle-Grid Integration Council (“VGIC”) hereby submits this response to the above-referenced Advice Letter 5536-E of Southern California Edison Company (“SCE”), *Southern California Edison’s Request for Approval to Update Its Low Carbon Fuel Standard Holdback Revenue Return Implementation Plan to Include a New Vehicle Grid Integration Program and to Qualify for Exemption from Public Utilities Code Section 851* (“Advice Letter”), submitted on April 29, 2025.

I. INTRODUCTION.

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of electric vehicles (“EV”) and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from EV deployments and flexible EV charging and discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid.

II. DISCUSSION.

VGIC commends SCE for proposing the Orchestrated Charging and Advanced Resiliency for Distribution (“ORCHARD”) Program and for pursuing Low Carbon Fuel Standard (“LCFS”) funding to support vehicle-grid integration (“VGI”).¹ The ORCHARD Program would apply VGI strategies to manage secondary local grid constraints, a growing concern as electrification increases strain on the distribution system that current time-of-use rates alone cannot sufficiently alleviate.

¹ See CARB’s recent LCFS Amendments, [Attachment A-1: Final Regulation Order \(Proposed Sections for Amendments\)](#) at p.49-50.

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By deploying VGI tools in ORCHARD, SCE can optimize the use of distribution assets, improve cost-effectiveness, and, if implemented correctly, defer or avoid costly infrastructure upgrades.

Notably, SCE's ORCHARD program design appears to directly respond to the California Air Resources Board's ("CARB") updated LCFS regulation, which explicitly details the use of LCFS holdback credit proceeds for VGI programs.² CARB provides the following examples of uses for these holdback credit proceeds:

- i. Investments in grid-side distribution infrastructure necessary for EV charging.
- ii. Support for vehicle-grid integration with projects such as:
 - I). Encouraging the optimization of EV charging through education in the following areas: peak demand, rate pricing, grid emergencies, potential power shutoffs, infrastructure deferral, renewable integration, and/or other signals and grid needs to provide grid and customer benefits.
 - II). Providing program incentives to encourage driver participation in monitored/managed charging, demand response, or vehicle-to-load / vehicle-to-grid applications.
 - III). Supporting the deployment and installation of bidirectional charging equipment.
 - IV). Other innovative approaches to promoting and managing EV charging and discharging that provides benefits to customers and the grid.
- iii. Hardware and software that decrease the cost of or avoid updates to infrastructure, including load management software or outlet splitting.

The Commission and utilities should seize this no-regrets opportunity to launch non-ratepayer-funded programs that deliver ratepayer benefits and generate critical data to inform the broader VGI market.

While VGIC offers recommendations for the implementation of ORCHARD, we do not believe that major Advice Letter modifications are required to address them. As such, VGIC strongly supports the timely approval of the implementation plan as filed, and encourages SCE to remain sufficiently flexible and responsive to customer and industry feedback during the implementation period.

A. VGIC Supports Approval of ORCHARD at Full Scale, Including Bidirectional Charging Incentives.

SCE proposes to use \$23 million in LCFS Holdback Funding to create ORCHARD and has an enrollment target of at least 25,000 customers. VGIC strongly supports this scale of deployment, which represents a meaningful step forward for VGI and provides an opportunity to demonstrate and maximize the cost-effectiveness of primarily software-based managed charging programs. As discussed by SCE, achieving this scale for a VGI program will "demonstrate that these technically-viable solutions generate significant return on investments."³ Enrolling tens of thousands of

² *Ibid.*

³ Advice Letter Appendix A at 2.

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customers in these programs is critical, as “the benefits that VGI can provide to the grid start to become material once significant numbers of customers are participating.”⁴ The costs associated with scaling these largely software-based programs are relatively low, and having large amounts of participation will maximize the cost-effectiveness of the investments SCE will make to create the program.

VGIC also strongly supports SCE’s plan to offer incentives for bidirectional charging, targeting 8,700 ORCHARD participants. Enabling bidirectional charging expands the grid services that can be provided by EVs and advances multiple state goals, including:

- Reducing costs to ratepayers, including both EV drivers and non-EV customers,
- Advancing the state’s transportation electrification goals by reducing the total cost of EV ownership,
- Supporting grid reliability, including during extreme heat events or other times of high stress on the grid,
- Supporting customer and community resiliency, including during Public Safety Power Shutoffs (“PSPS”) and other resilience events, and
- Fostering a growing market for VGI services, including job growth in manufacturing, field technicians, and related employment opportunities.

ORCHARD participants installing bidirectional equipment will be able to access an \$800 incentive to cover interconnection application costs. Offering an \$800 incentive to offset interconnection costs is critical to enabling bidirectional adoption, given that these fees greatly exceed the standard \$94 fee for similarly-sized rooftop solar customers (i.e., Net Energy Metering or Net Billing Tariff customers) – which may even use the same inverter model as bidirectional charging customers – and are a disproportionate burden for bidirectional charging systems. Unlike stationary storage, the EV battery providing bidirectional energy has already been purchased for another purpose, making interconnection costs one of the highest barriers to adoption (as opposed to the purchase price of the energy storage system). Given the current high interconnection costs and limited compensation mechanisms for EV battery discharge, many customers installing bidirectional equipment currently are doing so only for backup power. For clarity, VGIC notes that these backup-only system configurations do not currently require an \$800 interconnection fee and, instead, follow SCE’s longstanding safety rules applicable to the installation of customer-owned backup generators (i.e., SCE requests the customer notify the utility of the system’s existence and location, but the backup power system’s impact on the distribution grid does not need to go through an interconnection study process).

⁴ Advice Letter Appendix A at 2.

To achieve SCE's goal of enrolling 8,700 customers using grid-parallel bidirectional charging systems (as opposed to islanded, backup-only systems), customers must be able to access compensation mechanisms for grid-parallel EV battery discharge. Currently, SCE customers can receive compensation for EV exports during grid emergencies if eligible to participate in the Emergency Load Reduction Program ("ELRP") or Demand Side Grid Support ("DSGS") program. However, these programs cannot be relied on to continue in the long term⁵ and have funding that appears to be at risk.⁶ While customer export compensation that could apply to EV drivers was envisioned in Energy Division staff's *CalFUSE* dynamic pricing framework, SCE has elected not to offer export compensation to EV customers in its new Expanded Dynamic Rate Pilot.⁷ Lastly, SCE highlights its proposed Vehicle Grid Resource Proposal ("VGRP") as a potential EV export rate. VGIC encourages SCE and the Commission to move forward with establishing a static export compensation rate for EVs that recognizes the full value provided by customer exports, for example, as outlined in the Commission's adopted Avoided Cost Calculator.

If there is no compelling revenue stream for EV exports at the time of implementation, SCE should consider increasing the bidirectional charging customer incentive to achieve enrollment goals. VGIC supports SCE's plan to offer up to \$7,800 in additional incentives for income-qualified customers to help offset other equipment costs and ensure equitable access to bidirectional charging. Similar equipment cost offsets could be offered to general market customers, or other incentive structures could be explored. SCE should remain flexible and responsive to customer needs to increase incentive levels for bidirectional charging systems, if needed.

B. VGIC supports SCE's consideration that commercial fleets and medium- and heavy-duty vehicles should also be eligible to receive the bidirectional charging system incentive.

While SCE's proposal emphasizes residential enrollment, VGIC notes that SCE's materials reference both residential and commercial incentives as potentially eligible for bidirectional charging incentives. The Advice Letter outlines goals of enrolling residential customers and highlights the program's focus on mitigating distribution congestion arising from light-duty EV adoption.⁸ At the same time, the program documents note that bidirectional equipment incentives will be "available to both residential and commercial customers."

VGIC supports SCE's vision on how to incorporate commercial and medium- and heavy-duty (MDHD) into ORCHARD, specifically as it relates to bidirectional charging system incentives. These customers may have different transportation needs and charging schedules, but they could

⁵ For example, ELRP is only approved through 2027.

⁶ Governor Newsom's most recent May Revise budget has made cuts to funding that was previously allocated to DSGS.

⁷ See Public Advocates Office. *Prepared Testimony of Additional Rate Design Proposals in Southern California Edison's 2025 General Rate Case Phase 2*. November 22, 2024. Attachment B at 11-14. <https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2403019/7847/546561478.pdf>

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still provide significant value by participating in VGI initiatives – particularly given the large battery sizes and potential interest in unlocking additional value streams that reduce the total cost of EV ownership.

As SCE develops outreach and engagement strategies, including marketing and enrollment pathways, there may be opportunities to align ORCHARD with other commercial-focused initiatives, such as with the Charge Ready Programs, MDHD rebate programs, and MDHD financing programs offered by the utility. These touchpoints could support customer awareness and uptake among fleet operators and may inform future LCFS program proposals.

C. VGIC believes SCE should be provided ample flexibility in contracting with one or more aggregators in ORCHARD.

SCE indicates it will contract with at least one third-party aggregator and software provider to implement ORCHARD. VGIC encourages SCE to engage with the party or parties that are best suited to support customer outreach, program resilience, and interoperability concerns as revealed through common competitive procurement practices. In the Advice Letter, SCE outlines potential performance risks for ORCHARD, including “Aggregator software performance issues, such as a lack of interoperability or falling short in performance, will affect the reliability and efficacy of orchestrated charging.”⁹ By following long-established practices for developing partnership(s) with industry leaders, SCE can directly address this risk.

III. CONCLUSION.

VGIC appreciates the opportunity to submit this response to SCE’s Advice Letter. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Executive Director

VEHICLE-GRID INTEGRATION COUNCIL

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⁹ Advice Letter Appendix A at 28.